

Lab 6: Synchronous Counters

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Summary

This report examines the design, simulation, and hardware implementation of a 4-bit synchronous counter using JK flip-flops. Beginning with a required count sequence, the corresponding state diagram and state table were used to determine the excitation inputs of the flip-flops, and Karnaugh-map simplification was then used to obtain the Boolean expressions needed for the design. The circuit was implemented in Quartus, verified through simulation waveforms, and tested on the Altera board, where the counter outputs were displayed as binary values on LEDs. By comparing the theoretical sequence with the simulated and experimental results, this lab demonstrates the operation of synchronous counters and the importance of common-clock sequential design.

Introduction

Background

Counters are an important class of sequential logic circuits because their outputs depend on both the present inputs and the previously stored state of the system. In a synchronous counter, all flip-flops are driven by the same clock signal, which means that all state bits change together at the active clock edge. This makes synchronous counters more controlled and predictable than counters in which state changes ripple from one stage to the next. The starting point in the design of a synchronous counter is the count sequence, which may be represented by a state diagram or a state table. The count sequence is 0, 8, 12, 13, 15, 14, 6, 2, 3, 1, 0. Since these states are represented using 4 bits, four flip-flops are required for the implementation. The lab example makes use of JK flip-flops, which are commonly used in synchronous counter design.

Objective of the Report

The objective of this report is to document the design and testing of a synchronous counter and to compare the theoretical design procedure with the simulation and hardware results obtained in the lab. This includes deriving the state table from the given count sequence, obtaining the required JK inputs from the JK excitation table, simplifying the Boolean expressions, implementing the circuit in Quartus, and verifying the resulting counter sequence on the development board.

Equipment

Hardware

- Altera DE2-115 Development and Education Board
- USB-Blaster Programming Cable

Software

- Intel Quartus Prime
- Quartus Block Diagram/Schematic Editor
- Quartus Simulation Waveform Editor

Logic Components

- JK flip-flops
- AND, NOT, and other basic logic gates
- Common clock input

Input and Output Devices

- Toggle switches
- Push buttons or assigned clock input
- LEDs on the DE2-115 board for observing Q3, Q2, Q1, and Q0

Additional Equipment

- Computer
- USB connection cable

Design Procedure

The purpose of this experiment was to design, simulate, and test a 4-bit synchronous counter using JK flip-flops. The design process began with the count sequence provided in the lab instructions. From this sequence, a state diagram and state table were obtained. The JK excitation table was then used to determine the required inputs for each flip-flop, and the resulting expressions were simplified before the complete circuit was drawn in Quartus. After compilation and simulation, the counter was tested on the board and the observed output sequence was compared with the theoretical design.

Part 1 – Counter Design from the State Sequence

The first stage of the experiment involved analyzing the count sequence defined by the given state diagram. The sequence used for the synchronous counter was 0, 8, 12, 13, 15, 14, 6, 2, 3, 1, 0. A state table was then created by listing each present state together with its next state. Because the counter uses 4-bit states, four flip-flops were required. Any unused states were treated as don't-care conditions during the logic simplification stage. This step established the exact transitions that the synchronous counter needed to follow at each clock pulse.

State Table

Current (present) State				Next State				Q3 input		Q2 input		Q1 input		Q0 input	
Q3	Q2	Q1	Q0	Q3+	Q2+	Q1+	Q0+	J3	K3	J2	K2	J1	K1	J0	K0
0	0	0	0	1	0	0	0	1	X	0	X	0	X	0	X
0	0	0	1	0	0	0	0	0	X	0	X	0	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	0	0	1	0	X	0	X	X	1	X	0
0	1	1	0	0	0	1	0	0	X	X	1	X	0	0	X
1	0	0	0	1	1	0	0	X	0	1	X	0	X	0	X
1	1	0	0	1	1	0	1	X	0	X	0	0	X	1	X
1	1	0	1	1	1	1	1	X	0	X	0	1	X	X	0
1	1	1	0	0	1	1	0	X	1	X	0	X	0	0	X
1	1	1	1	1	1	1	0	X	0	X	0	X	0	X	1

Note:

All missing rows are don't cares

Part 2 – Derivation of JK Flip-Flop Inputs

Once the state table was completed, the excitation table of the JK flip-flop was applied to each state transition in order to determine the necessary J and K inputs for all four flip-flops. After this, K-maps were used to simplify the equations. The simplified Boolean expressions given in the instructions were:

$$J_0 = Q_1'Q_0'$$

$$K_0 = Q_0$$

$$J_1 = Q_2Q_0$$

$$K_1 = Q_0$$

$$J_2 = Q_3Q_1$$

$$K_2 = Q_1$$

$$J_3 = Q_0'$$

$$K_3 = Q_1$$

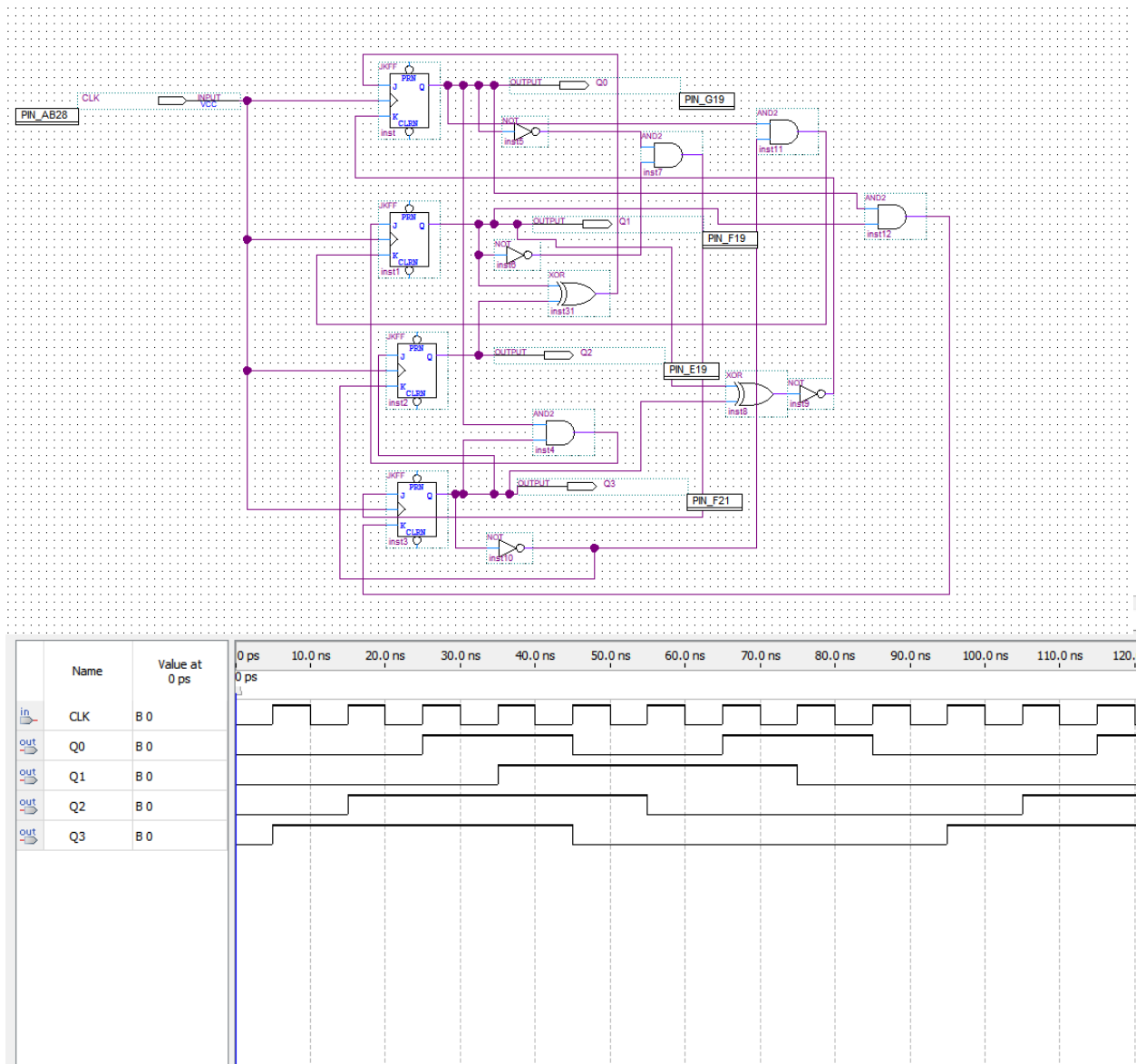
These expressions provided the combinational logic needed to drive the JK flip-flops so that the counter would move through the required sequence in a synchronous manner.

JK Flip-Flop excitation table:

Q(n)	Q(n+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Part 3 – Quartus Circuit Implementation and Simulation

A new project was created and a block diagram file was added to the project. The synchronous counter was then drawn using four JK flip-flops together with the logic derived from the simplified equations. Once the design was completed, the project was compiled to verify that there were no connection or synthesis errors. The circuit was then simulated using the waveform editor by applying clock pulses and observing the outputs Q3, Q2, Q1, and Q0. The waveform results were used to verify that the output states followed the intended count sequence at each active clock edge.



Part 4 – Experimental Verification on the DE2-115 Board

After the simulation matched the expected sequence, the circuit pins were assigned to the proper inputs and outputs on the Altera DE2-115 board. A clock source was used to advance the counter, and the four output bits were assigned to LEDs so that the state of the counter could be

observed directly in binary form. The experimental count table was then obtained by stepping through successive clock pulses and recording the LED outputs. These observations were compared with the theoretical state diagram and the simulation waveform to verify that the counter operated as intended.

Discussion of Results

The results obtained from simulation and hardware testing confirmed the expected behavior of the synchronous counter. The outputs changed only at the active clock event, which showed that all flip-flops were being controlled by the same clock and that the state transitions occurred simultaneously. This is the defining characteristic of a synchronous counter and demonstrates why this type of design is preferred when stable and predictable timing is required.

The observed sequence followed the state progression defined in the design. As the clock advanced, the counter moved through the required states and eventually returned to the initial state, confirming cyclic operation. The JK input equations derived from the state table successfully produced the desired next-state transitions, and the use of don't-care conditions simplified the combinational logic needed to implement the counter.

The waveform simulation provided a clear way to compare the theoretical and actual operation of the circuit. Each transition in the waveform corresponded to the intended count sequence, and the output lines Q3, Q2, Q1, and Q0 displayed the expected binary states. This showed that the logic equations were correctly implemented in Quartus and that the design responded properly to the applied clock signal.

The hardware implementation on the DE2-115 board further confirmed the correctness of the design. The LEDs displayed the current 4-bit state of the counter, allowing the sequence to be checked directly against the expected values from the state table. The experimental results matched the simulation results closely, showing strong agreement between theory, software verification, and physical implementation.

Overall, the lab demonstrated the full design process of a synchronous counter, from a given state sequence to a working hardware implementation. It reinforced the relationship between state diagrams, state tables, flip-flop excitation inputs, Boolean simplification, simulation, and real verification.

Conclusion

This lab focused on the design, simulation, and testing of a 4-bit synchronous counter using JK flip-flops. Starting from the count sequence provided in the instructions, the counter was developed by deriving the state table, applying the JK excitation table, and simplifying the resulting Boolean equations. The design was then implemented in Quartus, verified with simulation waveforms, and tested on the Altera development board using LEDs to display the binary counter outputs.

The results showed strong agreement between the theoretical sequence and the experimental observations. The counter advanced through the required states in a synchronous manner,

confirming that the flip-flop inputs and logic connections were correctly designed. This experiment therefore provided a practical demonstration of how synchronous counters are created and verified, and it reinforced the importance of common-clock sequential systems in digital circuit design.

Appendices

Appendix A - Synchronous Counters and JK Flip - Flop

- JK_ExcitationTable.png
- SynchronusCounterStateTable.png

Appendix B - Implementation of synchronous counter in Quartus Software

- synchronus_counter.bdf
- synchronus_counter.qsf
- synchronus_counter.qpf
- Waveform.vwf
- synchronusCounterCircuitDiagram.png
- synchronusCounterSimulation.png